

Prototype Methods aur `__proto__` ke bina objects

Modern tareeke

`obj.__proto__` purana hai. Behtar:

Kaam	Method
Prototype padhna	<code>Object.getPrototypeOf(obj)</code>
Prototype set karna	<code>Object.setPrototypeOf(obj, proto)</code>
Given prototype se naya object	<code>Object.create(proto, descriptors)</code>

Literal me `{ __proto__: animal }` likhna theek maana jaata hai.

javascript

```
let animal = { eats: true };
let rabbit = Object.create(animal); // {__proto__: animal} jaisa
Object.getPrototypeOf(rabbit) === animal; // true
```

`Object.create` ka doosra argument property descriptors leta hai:

javascript

```
let rabbit = Object.create(animal, { jumps: { value: true } });
```

Poora exact clone

javascript

```
let clone = Object.create(
  Object.getPrototypeOf(obj),
  Object.getOwnPropertyDescriptors(obj)
);
```

Isme saari properties (enumerable/non-enumerable, getters/setters) aur sahi prototype aa jaate hain.

Thoda itihaas

- Sabse purana: `F.prototype`.
- 2012: `Object.create`.

- 2015: `getPrototypeOf` / `setPrototypeOf` aaye. `__proto__` ko purana maana gaya.
- 2022: object literal me `__proto__` officially allowed, par `obj.__proto__` getter/setter abhi bhi purana hi hai.

Speed ka dhyan: existing object ka prototype baad me badalna bahut **slow** hota hai. Prototype ek baar creation ke waqt set karo.

"Bahut plain" objects (dictionary)

Agar user ki di hui keys object me store karte ho, to `"__proto__"` key ek problem hai. Kyunki `__proto__` sirf object/null accept karta hai, string assign karne par ignore ho jaata hai.

javascript

```
let obj = {};
obj["__proto__"] = "some value";
obj["__proto__"]; // [object Object], "some value" nahi!
```

`__proto__` asal me `Object.prototype` ka accessor hai, isliye ye hota hai.

Solutions:

1. `Map` use karo (sabse safe).
2. `Object.create(null)` se aisa object banao jiska prototype hi nahi:

javascript

```
let dict = Object.create(null);
dict["__proto__"] = "some value"; // ab normal key jaisa chalega
```

Aise object me `toString` jaise built-in methods nahi hote (ye aksar theek hai), lekin `Object.keys(dict)` jaise `Object.xyz()` functions chalte hain.

Task se seekhi baat

javascript

```
rabbit.sayHi(); // this = rabbit -> naam dikhega
Rabbit.prototype.sayHi(); // this = Rabbit.prototype -> undefined
Object.getPrototypeOf(rabbit).sayHi(); // undefined
```

`this` hamesha dot se pehle wala object hota hai.

Yaad rakho

- `Object.create`, `getPrototypeOf`, `setPrototypeOf` use karo.

- `Object.create(null)` = pure dictionary object.
- Prototype baar-baar mat badlo.